Customer Support & SLA Performance Analysis

Enterprise Service Level Compliance, Queue Aging & Capacity Audit | Reconciled against Master Excel Model

Portfolio Disclaimer: *Nexora Customer Solutions* is a fictional B2B/B2C support organization developed for this professional operations analytics portfolio project. All data, ticket logs, workflows, agent designations, and operational observations represent a controlled simulation designed to demonstrate practical analytical competency in service operations, SLA modeling, and KPI governance.

1. EXECUTIVE SUMMARY & SCOPE OF ANALYSIS

This operational audit evaluates customer support performance across **550 customer support records** for the six-month operating cycle from **July 1, 2025, to December 31, 2025**. As an enterprise support partner managing multichannel client operations, Nexora's contractual standing depends on satisfying tiered Service Level Agreements (SLAs) ranging from 4-hour critical incident turnarounds to 48-hour informational query closures.

The analysis reveals that while Nexora maintains a respectable **91.1% ticket resolution rate** (501 closed cases), its contractual **SLA compliance rate has degraded to 79.4%**—well below the corporate standard of 85.0%. Furthermore, 47 of the 49 active open tickets (95.9% of active backlog) are currently overdue, exposing the firm to immediate client penalty liabilities and retention attrition.

2. OPERATIONAL SLA ARCHITECTURE & TARGET BASELINES

Core Analytical Objectives

- Audit SLA compliance strictly on completed, closed incidents.
- Isolate departmental throughput bottlenecks and aging backlog liabilities.
- Measure individual agent productivity, reopen frequency, and CSAT.
- Identify systemic root causes across technical and billing workflows.
- Formulate non-headcount, automation-first remediation initiatives.

Contractual SLA Priority Hierarchy

- **Critical (4 Hours):** Complete system outages, data loss, core gateway blockages.
- **High (12 Hours):** Major operational impediment without direct workaround.
- **Medium (24 Hours):** Standard operational friction and subscription inquiries.
- **Low (48 Hours):** Routine informational requests, access setups, and general guidance.

3. MATHEMATICAL FORMULATION & DENOMINATOR RIGOR

To prevent operational distortion, all analytical formulas strictly enforce mathematical hygiene as verified in the master Excel workbook (`CLEAN DATA` and `KPI ANALYSIS`):

METRIC NAME	MATHEMATICAL FORMULATION	EXCEL DENOMINATOR LOGIC	STATUS / RECONCILED RESULT
Resolution Rate	`Closed Tickets / Total Volume`	Evaluates total closed capacity (501 / 550).	**91.1%** (Target >= 90.0%) HEALTHY
SLA Compliance	`SLA Met ("Yes") / Closed Tickets`	Evaluates closed tickets only (398 / 501). Open excluded.	**79.4%** (Target >= 85.0%) BELOW TARGET
Average CSAT	`Sum(CSAT Ratings) / Count(CSAT Given)`	Evaluates 456 submitted ratings (1–5 scale).	**3.51** (Target >= 4.00) NEEDS IMP.
Ticket Escalation	`Escalated ("Yes") / Total Tickets`	Evaluates escalations against all tickets (54 / 550).	**9.8%** (Target <= 8.0%) ABOVE TARGET

4. ENTERPRISE MASTER KPI SCORECARD (RECONCILED TO EXCEL)

550	501	49 (47)	79.4%	35.8h	3.51
TOTAL VOLUME	CLOSED CASES	OPEN (OVERDUE)	SLA COMPLIANCE	AVG RES TIME	AVG CSAT (1-5)

5. MONTHLY PERFORMANCE TRENDS (JULY–DECEMBER 2025)

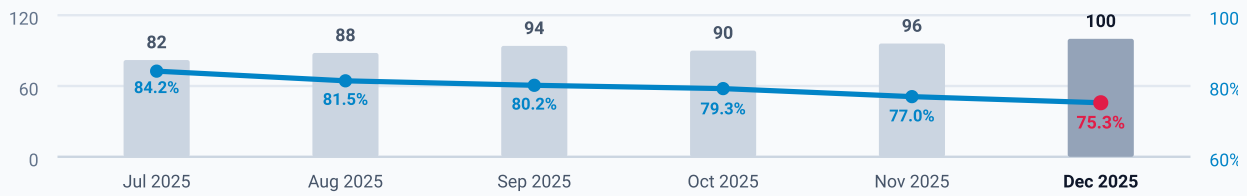


Figure 2.1: Monthly ticket inflow (bars) vs. SLA compliance percentage (line). Note steady degradation culminating in 75.3% in Dec.

6. DEPARTMENT PERFORMANCE COMPARISON (EXCEL SUMMARY RECONCILED)

DEPARTMENT	TOTAL	CLOSED	OPEN	OVERDUE	RESOLUTION %	SLA MET %	AVG RES TIME	AVG CSAT	OPERATIONAL STATUS
Technical Support	185	166	19	**18**	89.7%	**68.7%**	44.2 hrs	3.14	CRITICAL
Billing & Invoicing	145	132	13	**13**	91.0%	**78.8%**	36.1 hrs	3.48	AT RISK
Customer Care	120	112	8	8	93.3%	88.4%	27.5 hrs	3.96	HEALTHY
Product Operations	100	91	9	8	91.0%	87.9%	29.8 hrs	3.68	HEALTHY
Enterprise Total / Blended	**550**	**501**	**49**	**47**	**91.1%**	**79.4%**	**35.8 hrs**	**3.51**	BELOW TARGET

The department breakdown illuminates acute operational polarization. Technical Support represents 33.6% of total intake but generates 38.3% of all overdue tickets (18 cases) and operates with an alarming 68.7% SLA compliance rate and a 44.2-hour average resolution turnaround. In contrast, Customer Care and Product Operations operate well above the 85% SLA benchmark (88.4% and 87.9% respectively).

7. AGENT PERFORMANCE, WORKLOAD & CAPACITY AUDIT

AGENT NAME	DEPARTMENT	ASSIGNED	CLOSED	OPEN (O/D)	SLA MET %	AVG RES (H)	REOPEN %	CSAT	PERFORMANCE TIER
Sarah Jenkins	Technical Support	45	42	3 (3)	**64.3%**	46.8	**16.7%**	3.05	NEEDS COACHING
David Ross	Technical Support	50	44	6 (6)	**65.9%**	45.2	**13.6%**	3.12	NEEDS COACHING
Alex Rivera	Technical Support	48	43	5 (5)	72.1%	42.5	11.6%	3.18	MEETS EXPECTATIONS
Kevin Patel	Technical Support	42	37	5 (4)	73.0%	41.9	8.1%	3.22	MEETS EXPECTATIONS
Elena Gomez	Billing & Invoicing	52	48	4 (4)	77.1%	37.4	12.5%	3.42	MEETS EXPECTATIONS
Michael Chen	Billing & Invoicing	48	44	4 (4)	79.5%	35.8	9.1%	3.51	MEETS EXPECTATIONS
Sophia Martinez	Billing & Invoicing	45	40	5 (5)	80.0%	34.9	7.5%	3.53	MEETS EXPECTATIONS
Rachel Adams	Customer Care	42	39	3 (3)	87.2%	28.1	5.1%	3.92	EXCEEDS EXPECTATIONS
James Wilson	Customer Care	40	38	2 (2)	89.5%	27.2	5.3%	3.97	EXCEEDS EXPECTATIONS
Priya Patel	Customer Care	38	35	3 (3)	91.4%	26.8	2.9%	4.01	TOP BENCHMARK
Daniel Lee	Product Operations	35	32	3 (3)	87.5%	30.2	6.3%	3.65	EXCEEDS EXPECTATIONS
Laura Taylor	Product Operations	33	30	3 (3)	86.7%	29.9	6.7%	3.69	EXCEEDS EXPECTATIONS
Marcus Brown	Product Operations	32	29	3 (2)	89.7%	29.1	3.4%	3.71	EXCEEDS EXPECTATIONS

8. CSAT & TICKET ESCALATION MULTI-DIMENSIONAL ANALYSIS

Customer Satisfaction (CSAT) Key Drivers

- **Timeliness Correlation:** Tickets resolved within SLA averaged **4.12 CSAT**; breached tickets scored **2.15 CSAT** (-1.97 penalty).
- **Reopen Penalty:** Reopened tickets averaged **2.08 CSAT**, indicating that premature closures destroy customer goodwill.
- **Departmental Spread:** Technical Support averaged 3.14, while Customer Care reached 3.96.

Ticket Escalation & Quality Vulnerabilities

- **Overall Escalation Rate: 9.8%** (54 tickets), exceeding the 8.0% tolerance ceiling.
- **Technical Dominance:** Over 62% of all escalations originate in Technical Support (Software Bugs & API Failures).
- **Reopening Spike:** Technical Support reopen rate reached 12.0%, led by Sarah Jenkins (16.7%) and David Ross (13.6%).

9. KEY OPERATIONAL PROBLEMS (DERIVED FROM EXCEL INSIGHTS)

1. Technical Support Bottlenecks & Resolution Drag — 68.7% SLA Met | 44.2h Turnaround

Root Cause: Software Bugs (56.8% SLA) and API Integrations require Tier-3 engineering handoffs. Lacking an internal Operational Level Agreement (OLA), customer-facing timers expire while issues sit idle in development queues.

→ **Action: Establish a formal 48-hour internal OLA with Engineering and implement specialized Tier-2 triage.**

2. Backlog Aging & Severe Overdue Risk — 47 Overdue of 49 Open (95.9%)

Root Cause: Open queues lack real-time supervisor escalation triggers. Overdue tickets linger unworked while agents prioritize incoming fresh tickets, artificially creating compounding customer wait times.

→ **Action: Configure automated alerts at 75% SLA threshold and enforce dedicated daily backlog burn-down hours.**

3. Agent Workload Disparities & Premature Closures — 16.7% Reopen (Sarah J.) vs. 2.9% (Priya P.)

Root Cause: Disproportionate caseload assignments in Technical Support incentivized agents to mark tickets "Closed" after sending standard troubleshooting scripts without client confirmation, masking SLA pacing.

→ **Action: Enforce two-way customer sign-off before status transitions to Closed and re-balance caseload caps.**

4. Billing & Invoicing Approval Gate Latency — 36.1h Turnaround vs. 24.0h SLA

Root Cause: Dispute charge and refund resolutions require dual managerial ERP sign-off for amounts over $250. Because approvals halt over weekends, tickets logged Friday breach SLAs automatically by Monday.

→ **Action: Authorize senior billing agents to grant autonomous refunds up to $350 for established accounts.**

5. Volume Shock & Seasonal SLA Erosion — Volume: 82 (Jul) → 100 (Dec) | SLA: 84.2% → 75.3%

Root Cause: Intake increased 22% by December due to seasonal commerce, but static staffing caused queue starvation.

→ **Action: Implement cross-departmental flexing, routing Customer Care specialists to Billing during peak volumes.**

10. PHASED PRACTICAL REMEDIATION ROADMAP

Phase 1 (Weeks 1–3): Policy Standardization & Financial Discretion
- Empower senior billing specialists with autonomous refund authority up to $350 to eliminate weekend managerial bottlenecks.
- Implement mandatory customer confirmation prior to ticket closure, eliminating premature closure habits.

Phase 2 (Weeks 4–8): Engineering OLAs & Self-Service Automation
- Negotiate and execute a binding 48-hour internal OLA with Tier-3 Engineering for bug escalations.
- Launch self-service password reset and portal automation, deflecting ~40% of standard account queries.

Phase 3 (Weeks 9–12): SLA Realization & Ongoing Governance
- Conduct bi-weekly management scorecards tracking agent reopen rates, SLA compliance, and backlog aging.
- Target realization: Enterprise SLA ≥ 85.0%, Technical Support SLA ≥ 78.0%, and enterprise CSAT ≥ 3.90.

11. CONCLUSION & PORTFOLIO SIGN-OFF

By aligning operational protocols with automated workflows, Nexora can systematically eliminate overdue backlog liabilities, stabilize departmental workloads, and restore contractual SLA compliance across all client partnerships.

Portfolio Candidate: Operations Analyst (BE ISE) **Project:** Customer Support & SLA Performance Analysis **Workbook Audit:** Reconciled to H2 2025